DESCRIPTIVE STATISTICS · STUDY NOTES

Covariance

Do two variables move together, and in which direction?

30-SECOND REVISION

Covariance measures how two variables move together. Multiply each pair's distances from their means, add them up, and divide by **n − 1** (sample) or **N** (population).
Sign tells direction: positive = move together, negative = move in opposite directions, zero = no linear relationship.
Weakness: the size of the number is hard to interpret (house data: **33,491,250**). The fix is the **correlation coefficient**, the next topic.

Contents: 1 Where this fits · 2 What covariance is · 3 Worked example with the housing data · 4 Reading the number · 5 Exercise: SAT scores · 6 Excel and Python · 7 Revision kit · 8 Notes check

1. Where this fits

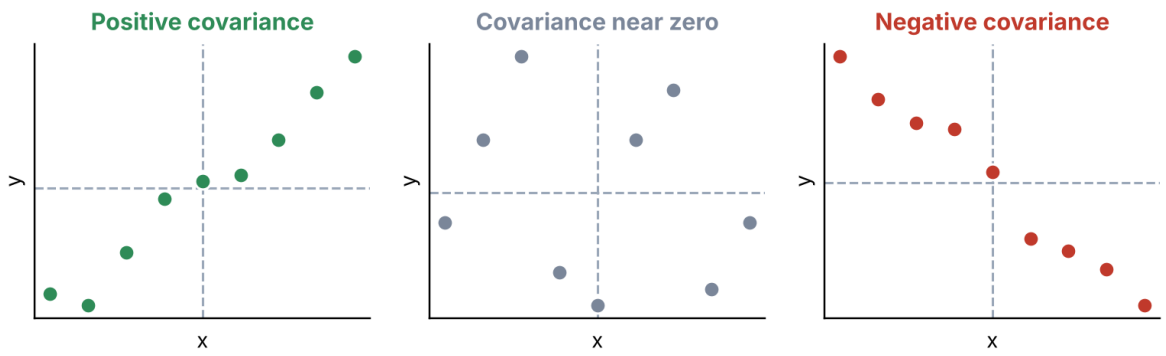
So far every measure described **one variable** at a time. Now the class moves to measures that work with **more than one variable**. The two main ones are covariance and the linear correlation coefficient.

Kind	Measures	Describes
Univariate	Mean, median, mode · skewness · variance, standard deviation, coefficient of variation	One variable
Bivariate	**Covariance** · correlation coefficient	**The relationship between two variables**

The running example is real estate: **what determines house prices?** Larger houses tend to cost more because people like more space, so there are two variables, **size** and **price**. A scatter plot (size on the x-axis, price on the y-axis) shows a clear relationship, and covariance is the statistic that measures it.

2. What covariance is

Definition. Covariance measures how two variables vary together. It can be positive, zero or negative, and its **sign** gives the direction of the relationship.



Illustrative patterns (not from the class data). Dashed lines mark each variable's mean.

Covariance	Meaning	Example
> 0 (positive)	The two variables move **together**	Bigger house, higher price
< 0 (negative)	The two variables move in **opposite directions**	Ice-cream sales and umbrella sales
= 0	No linear relationship; the variables move independently of each other	Unrelated quantities

The formulas

$$\operatorname{cov}(x, y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

$$\operatorname{cov}(x, y) = \frac{\sum_{i=1}^{N} (x_i - \mu_x)(y_i - \mu_y)}{N}$$

	Sample	Population
Size	n	N
Means	$\bar{x}$ and $\bar{y}$	μ_x and μ_y
Denominator	n − 1	N

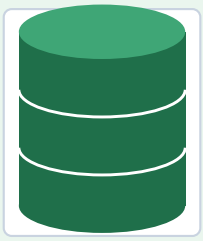
Why multiply the two differences?

For every observation, take its distance from the mean of x and its distance from the mean of y, and multiply them. If both are above their means (or both below), the product is **positive**. If one is above and the other below, the product is **negative**. Adding the products shows which kind of pair dominates: mostly positive products give a positive covariance, mostly negative products give a negative one.

3. Worked example: housing data

Five houses, with their size and price. Because these five houses are only a part of all houses, this is a **sample**, so the sample formula (÷ n − 1) is used.

3.1 The data set



click to open

Housing: Size vs Price

DATA SET
EXAMPLE

Data set example · 24.9 KB · 3 sheets

Housing data: 5 houses, size and price, with covariance worked out in Excel

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

House	Size (ft.)	Price ($)
1	650	$ 772,000
2	785	$ 998,000
3	1,200	$ 1,200,000
4	720	$ 800,000
5	975	$ 895,000

3.2 The steps

Step 1: means. Mean size = 4,330 ÷ 5 = **866** ft. Mean price = 4,665,000 ÷ 5 = **$ 933,000**.

Step 2: distances and products. For each house, multiply (size − mean size) by (price − mean price). The table below shows all five.

Step 3: add the products. Sum of the five products = **133,965,000**.

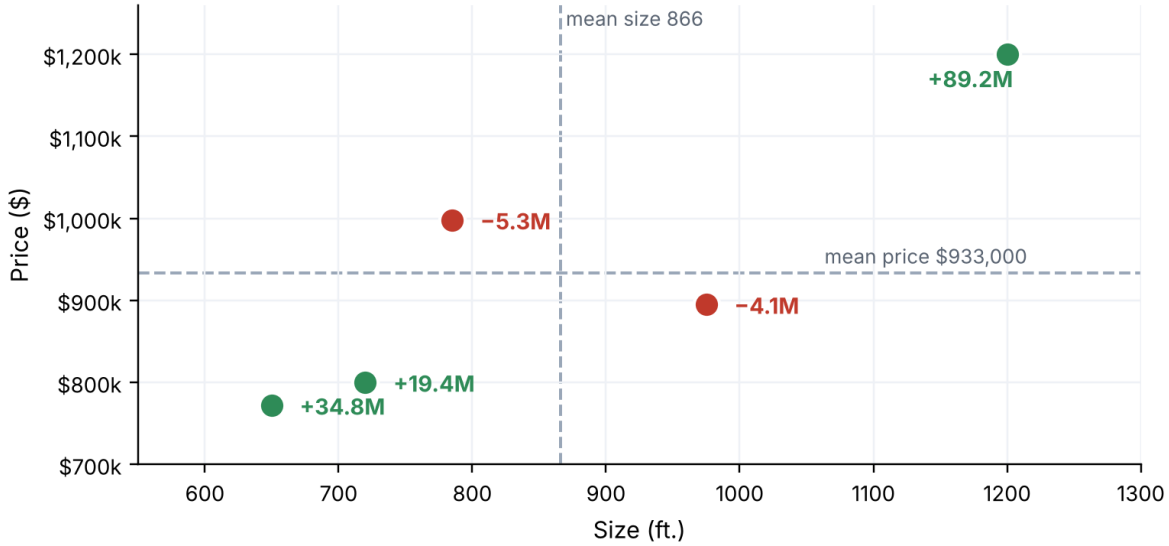
Step 4: divide by n − 1. The sample size is 5, so divide by 4: 133,965,000 ÷ 4 = **33,491,250**.

Distances from the mean and their products

House	x − 866	y − 933,000	(x − x̄)(y − ȳ)	Sign
1	-216	-161,000	+34,776,000	+
2	-81	+65,000	-5,265,000	−
3	+334	+267,000	+89,178,000	+
4	-146	-133,000	+19,418,000	+
5	+109	-38,000	-4,142,000	−
Sum			**133,965,000**	

$$\text{cov} = \frac{133{,}965{,}000}{5-1} = 33{,}491{,}250$$

Reading the scatter plot



Each point shows its contribution to the sum, in millions. Three houses contribute positive products and two contribute small negative ones, so the total is large and positive.

Result
Covariance = **33,491,250**. It is **positive**, so house size and price **move together**: larger houses tend to cost more. That is the “sense of direction” the class notes refer to.

4. Reading the number

- **The sign is meaningful.** It tells you the direction (together, opposite, or neither).
- **The size is hard to interpret.** A covariance could be 5, 50, 0.0023456 or 33,491,250. Is that strong or weak? You cannot tell.
- **Why?** Covariance is in the units of x multiplied by the units of y (here, size × price). Change the units and the number changes: measure price in thousands of dollars and the same data gives 33,491.25, although the relationship is identical.
- **Symmetric:** cov(x, y) = cov(y, x). Swapping the two variables gives the same value.

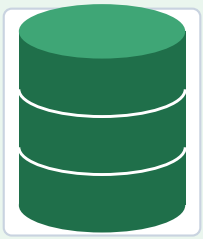
The fix: the correlation coefficient
To get a number that is easy to interpret, divide the covariance by the standard deviations of both variables. That gives the **correlation coefficient r**, which always lies between −1 and 1. It is the next topic in the class.

5. Exercise: SAT scores

From the exercise workbook. The background: you are given the SAT **reading** and **writing** scores of several students (data from the earlier lesson on cross tables and scatter plots).

Task	What to do	Answer
1	Determine whether this is a sample or a population	**Sample** (only 5 students)
2	Calculate the covariance of the two datasets	**21,155.55**
3	Plot a scatter plot and comment on any relationship	**Noticeable relationship** (positive)

5.1 The data set



click to open

SAT Scores: Writing vs Reading

DATA SET
EXAMPLE

Data set example · 29.5 KB · 3 sheets

Exercise and solution: 5 students, SAT writing and reading scores

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

Student	Writing (x)	Reading (y)
1	344	378
2	383	349
3	611	503
4	713	719
5	536	503

5.2 Solution, step by step

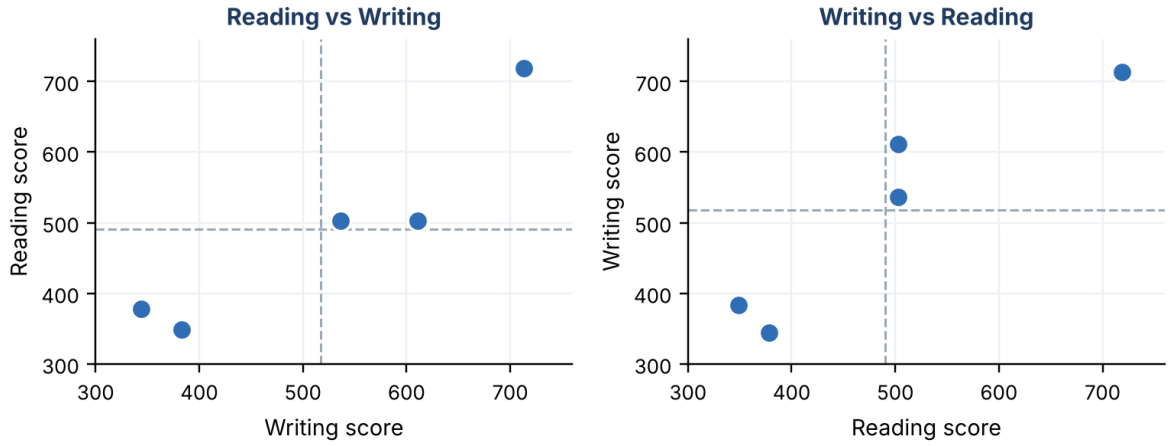
Task 1. Only five students were measured, which is obviously a small part of all students, so this is a **sample**. Use ÷ (n − 1).

Task 2. Mean writing = 2,587 ÷ 5 = **517.4**. Mean reading = 2,452 ÷ 5 = **490.4**. Multiply the two distances for each student, add them (84,622.20), then divide by 4.

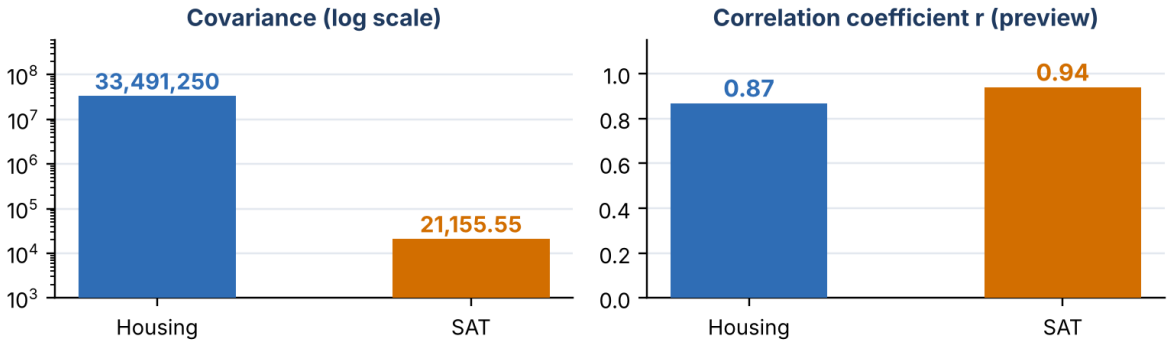
Student	x − 517.4	y − 490.4	$(x - \bar{x})(y - \bar{y})$
1	-173.4	-112.4	19,490.16
2	-134.4	-141.4	19,004.16
3	+93.6	+12.6	1,179.36
4	+195.6	+228.6	44,714.16
5	+18.6	+12.6	234.36
Sum			**84,622.20**

$$\text{cov} = \frac{84{,}622.2}{5-1} = 21{,}155.55$$

Task 3. The solution plots the data twice, once with each variable on the x-axis. The points rise from bottom left to top right, so there is a **noticeable positive relationship**: students with higher writing scores tend to have higher reading scores. This agrees with the positive covariance.



Why covariance alone can mislead (added preview)
The SAT covariance (**21,155.55**) is far smaller than the housing covariance (**33,491,250**), yet the SAT relationship is the **stronger** one (correlation r ≈ **0.94** vs **0.87** for housing). The two numbers cannot be compared because they are in different units. That is exactly why the correlation coefficient is needed.



Correlation values are from my own calculation and preview the next topic; they are not from the covariance class notes.

6. Excel and Python

Goal	Excel	Result (housing)
Sample covariance (÷ n − 1)	=COVARIANCE.S(C6:C10, D6:D10)	33,491,250
Population covariance (÷ N)	=COVARIANCE.P(C6:C10, D6:D10)	26,793,000
Built by hand, as in the workbook	=SUM(G6:G10)/(COUNT(C6:C10)-1)	33,491,250
One product term	=(C6-C11)*(D6-D11)	34,776,000

```
import statistics as st
import numpy as np

size  = [650, 785, 1200, 720, 975]
price = [772000, 998000, 1200000, 800000, 895000]

st.covariance(size, price)           # 33491250.0  (sample, n - 1)
np.cov(size, price)[0, 1]            # 33491250.0  (sample: ddof=1 by default)
np.cov(size, price, ddof=0)[0, 1]    # 26793000.0  (population)
```

Pitfall
`np.cov` returns a **2 × 2 matrix** (covariances of every pair), so pick the off-diagonal entry `[0, 1]`. Unlike `np.var`, it divides by n − 1 by default. In pandas, `Series.cov()` also divides by n − 1.

7. Revision kit

Formula sheet

Quantity	Sample	Population
Mean of x	$\bar{x} = \Sigma x \div n$	$\mu_x = \Sigma x \div N$
Covariance	$\Sigma(x - \bar{x})(y - \bar{y}) \div (n - 1)$	$\Sigma(x - \mu_x)(y - \mu_y) \div N$
Sign rule	> 0 together · < 0 opposite · = 0 no linear link	same

Self-test (cover the answers and try first)

#	Question	Answer
1	What do a positive, negative and zero covariance mean?	Positive: the variables move together. Negative: they move in opposite directions. Zero: no linear relationship.
2	Why is the sample formula used for the housing data?	The five houses are only a sample of all houses, so divide by n − 1 = 4.
3	What are the units of covariance?	The units of x times the units of y (for the housing data, size × price).
4	Data: x = 1, 2, 3 and y = 2, 4, 6. Find the sample covariance.	Means 2 and 4. Products: 2, 0, 2, sum 4. Divide by 2: **2** (positive).
5	The housing prices are changed from dollars to thousands of dollars. What happens to the covariance?	It is divided by 1,000 (33,491,250 becomes 33,491.25). The relationship has not changed.
6	Is cov(x, y) different from cov(y, x)?	No. They are equal.
7	Why can't you judge the strength of a relationship from the covariance alone?	Its size depends on the units. Use the correlation coefficient instead.

Common mistakes

- Dividing by n instead of **n − 1** when the data is a sample.
- Forgetting to subtract the means before multiplying.
- Reading a large covariance as a strong relationship, or a small one as a weak relationship.
- Treating a positive covariance as proof that one variable **causes** the other.

- Assuming covariance = 0 means the variables are completely unrelated (it only rules out a *linear* relationship).

8. Notes check: points to double-check

While preparing these notes I compared the class notes with the two workbooks. These small points are worth correcting in your own copy; the numbers in this document follow the workbooks.

Where	What I noticed	Suggested fix
Class notes, zero covariance	"=0 … move independent (house size & price both are equal)" is unclear.	Say: covariance = 0 means **no linear relationship**. Independent variables have zero covariance, but zero covariance does not by itself prove independence.
Workbook labels	The size column is labelled "Size (ft.)".	Housing size is normally in **square feet** (ft^2). Relabel it if that is what the data means.
Sheets "cov" and "Covariance2"	The sample size is typed in as 5 and the formula divides by a typed 4.	Use `=COUNT(...)` and `/(n−1)` as the "Covariance" sheet does, so they update with the data.
Exercise workbook, cell M12	The answer 21,155.55 is typed in.	Link it to the calculated cell G20 (`=G20`).

In one line each

- **Covariance** = $\Sigma(x - \bar{x})(y - \bar{y}) \div (n - 1)$, the direction in which two variables move together.
- Sign: **> 0** together, **< 0** opposite, **= 0** no linear link. Size: hard to interpret.
- Housing: cov = **33,491,250**. SAT exercise: cov = **21,155.55**. Both positive.

Source: class notes (Part 5), *02_Covariance.xlsx* (attached on page 3) and *03_Covariance_Exercise_Solution.xlsx* (attached on page 5). The pattern diagram, the sections on reading the number, the comparison with correlation, the revision kit and the notes check were added for clarity and are marked as additions. All numbers were checked by computation against the workbooks.